Development Phase Plan

Project Name

CarbonWise

Version: 1.0

Status: Approved

Methodology: Agile + Iterative Development

Duration: 8 Weeks

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1. Purpose

This document defines the complete development roadmap for CarbonWise.

The goal is to ensure development remains focused on the challenge statement:

> Design a solution that helps individuals understand, track, and reduce their carbon footprint through simple actions and personalized insights.

while maximizing performance in:

- Code Quality (High Impact)
- Problem Statement Alignment (High Impact)
- Security (Medium Impact)
- Efficiency (Medium Impact)
- Testing (Low Impact)
- Accessibility (Low Impact)

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2. Development Philosophy

CarbonWise will not be developed as a simple carbon calculator.

Instead, development focuses on:

```
Understand

↓

Track

↓

Reduce

↓

Behavior Change
```

This flow guides every engineering decision.

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3. Development Methodology

Framework

```
Agile Scrum
```

Sprint Duration

```
1 Week
```

Total Sprints

```
8 Sprints
```

Workflow

```
Planning
```

```
↓

Development

↓

Testing

↓

Review

↓

Deployment
```

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4. Phase 0 — Research and Planning

Objectives

Understand:

- Problem Statement
- User Personas
- Sustainability Challenges
- Carbon Tracking Methods

Deliverables

Product

- PRD
- User Personas
- User Stories

Architecture

- System Architecture

- Database Design
- API Contracts

Development Setup

- Git Repository
- Monorepo Structure
- CI/CD Pipeline

Success Criteria

✅ Requirements Finalized

✅ Architecture Approved

✅ Development Environment Ready

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5. Phase 1 — Core Foundation

Objectives

Build the platform foundation.

Features

Authentication

- Registration
- Login
- JWT Authentication
- Password Reset

User Profile

- Personal Information

- Sustainability Preferences

Initial Dashboard

- Basic Layout
- User Overview

Engineering Deliverables

Frontend:

```
Next.js Setup

Tailwind Setup

Shadcn Setup
```

Backend:

```
NestJS Setup

Prisma Setup

PostgreSQL Setup
```

Security:

```
Helmet

bcrypt

JWT

Rate Limiting
```

Success Criteria

✅ Users Can Register

✅ Users Can Login

✅ Security Layer Implemented

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6. Phase 2 — Carbon Intelligence Engine

Objectives

Implement carbon footprint understanding.

Features

Carbon Calculator

Categories:

```
Transportation

Food

Energy

Shopping
```

Carbon Score Engine

Features:

- Score Calculation
- Sustainability Rating

Activity Logging

Features:

- Add Activities
- Store Activities
- Retrieve Activities

Deliverables

Backend:

```
Carbon Engine

Emission Factors

Calculation Services
```

Frontend:

```
Calculator UI

Results Dashboard
```

Success Criteria

✅ Carbon Calculations Accurate

✅ Carbon Score Generated

✅ Users Understand Emissions

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7. Phase 3 — Tracking Engine

Objectives

Enable users to track sustainability progress.

Features

Dashboard Analytics

Displays:

- Monthly Trends
- Emission Breakdown
- Carbon History

Activity History

Features:

- Search
- Filtering
- Pagination

Reports

Features:

- Weekly Summary
- Monthly Summary

Deliverables

Analytics Engine

Dashboard Components

History Module

Success Criteria

✅ Users Can Track Progress

✅ Trends Display Correctly

✅ Dashboard Loads Under 2 Seconds

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8. Phase 4 — Recommendation Engine

Objectives

Enable personalized carbon reduction.

Features

Hotspot Detection

Identify:

```
Highest Emission Category
```

Recommendation Ranking

Based On:

```
Impact

Feasibility

User Preferences
```

Personalized Insights

Examples:

Transportation Heavy User:

```
Use Public Transport
```

Food Heavy User:

```
Reduce Meat Consumption
```

Deliverables

Recommendation Service

Insight Engine

Priority Engine

Success Criteria

✅ Recommendations Personalized

✅ Hotspots Identified Correctly

✅ Users Receive Actionable Advice

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9. Phase 5 — Goal Tracking System

Objectives

Create measurable sustainability improvement.

Features

Goal Creation

Users Can:

- Create Goals
- Edit Goals
- Delete Goals

Progress Tracking

Displays:

- Completion Percentage
- Milestones
- Target Reduction

Goal Analytics

Displays:

- Goal Success Rate
- Progress History

Deliverables

Goal Engine

Milestone System

Progress Calculator

Success Criteria

✅ Goals Created Successfully

✅ Progress Calculated Correctly

✅ Milestones Trigger Properly

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10. Phase 6 — Carbon Twin Engine

Objectives

Deliver predictive sustainability insights.

Why Important

This feature provides the strongest differentiation from traditional carbon calculators.

Features

Scenario Builder

Examples:

```
Reduce Car Usage

Use Public Transport

Reduce Meat Consumption

Lower Electricity Usage
```

Emission Forecasting

Displays:

```
Current Emissions
```

```
↓

Projected Emissions

↓

Potential Savings
```

Impact Visualization

Displays:

- Predicted Carbon Score
- Future Reduction
- Sustainability Forecast

Deliverables

Carbon Twin Engine

Simulation Logic

Forecast Dashboard

Success Criteria

✅ Scenarios Simulated Correctly

✅ Forecasts Generated

✅ Personalized Insights Improved

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11. Phase 7 — Testing and Quality Assurance

Objectives

Prepare for production release.

Testing Scope

Unit Tests

Technology:

```
Jest
```

Coverage Target:

```
90%+
```

Integration Tests

Technology:

```
Supertest
```

Coverage Target:

```
80%+
```

End-to-End Tests

Technology:

```
Playwright
```

Critical Flows:

- Registration
- Login
- Carbon Calculator
- Recommendations
- Goals
- Carbon Twin

Accessibility Testing

Tools:

```
Playwright

axe-core
```

Security Testing

Verify:

- JWT Security
- Helmet Headers
- Input Validation
- Rate Limiting
- Authorization

Performance Testing

Targets:

```
API < 500ms

Calculator < 100ms

Dashboard < 2s
```

Success Criteria

✅ All Tests Pass

✅ No Critical Bugs

✅ Security Requirements Met

✅ Accessibility Requirements Met

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12. CI/CD Integration Timeline

Implemented From:

```
Sprint 1
```

Pipeline:

```
Git Push

↓

ESLint

↓

Type Check

↓

Jest

↓

Supertest

↓

Build
```

```
↓

Playwright

↓

Deploy
```

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13. Security Roadmap

Implemented Throughout Development.

Sprint 1

Authentication Security

Sprint 2

Input Validation

Sprint 3

Rate Limiting

Sprint 4

Authorization Review

Sprint 5

Security Testing

Sprint 6

Dependency Audit

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14. Accessibility Roadmap

Implemented Throughout Development.

Navigation

Keyboard Accessible

Forms

Accessible Labels

Charts

Screen Reader Support

Color System

WCAG 2.2 AA Compliance

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15. Performance Roadmap

Implemented Throughout Development.

Database

Indexes

Pagination

Backend

Caching

Optimized Queries

Frontend

Lazy Loading

Code Splitting

Image Optimization

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16. Release Readiness Checklist

Before Release:

✅ Code Review Complete

✅ ESLint Passes

✅ Type Checking Passes

✅ Jest Passes

✅ Supertest Passes

✅ Playwright Passes

✅ Security Review Complete

✅ Accessibility Review Complete

✅ Performance Targets Met

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17. Risk Management

Risk 1

Incorrect Carbon Calculations

Mitigation:

- Verified Emission Factors
- Unit Testing

Risk 2

Poor Recommendation Quality

Mitigation:

- Hotspot Analysis
- Recommendation Ranking

Risk 3

Performance Bottlenecks

Mitigation:

- Redis
- Pagination
- Indexes

Risk 4

Security Vulnerabilities

Mitigation:

- Helmet
- JWT
- bcrypt
- Validation

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18. Final Success Criteria

The development roadmap is successful when:

✅ Users Understand Their Carbon Footprint

✅ Users Track Sustainability Progress

✅ Users Receive Personalized Insights

✅ Users Reduce Emissions Through Actions

✅ Code Quality Remains High

✅ Security Best Practices Are Followed

✅ Performance Targets Are Met

✅ Accessibility Requirements Are Satisfied

✅ The Platform Fully Solves The Challenge Statement